Economics 450-1, Problem Set 1: Figures and Tables

Question 6: market simulation and welfare

6a

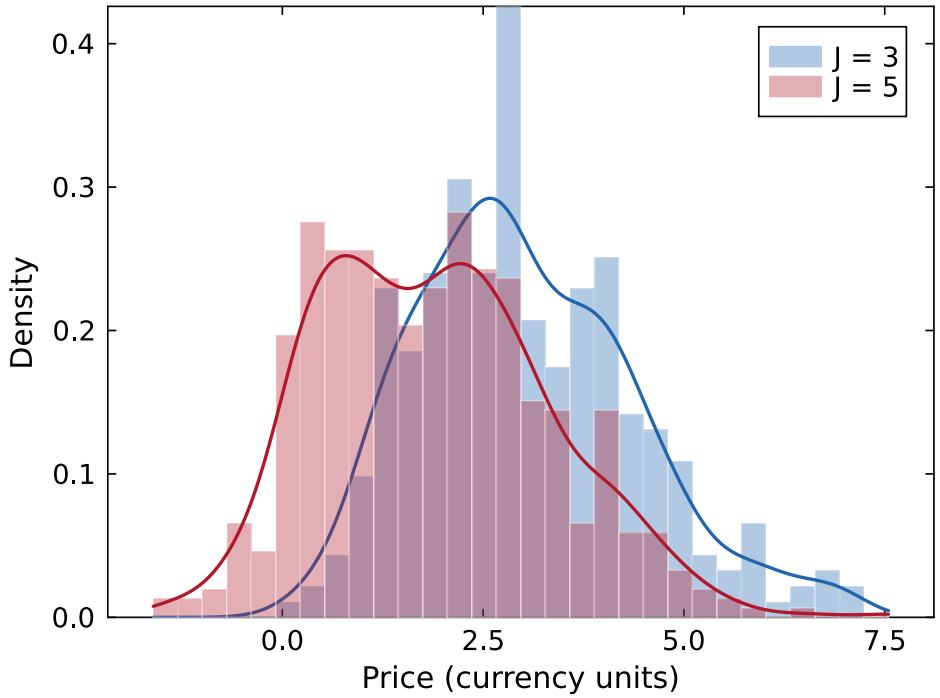


Figure 1: Distribution of prices, $J = 3$ versus $J = 5$.

Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` (`pset_io`, Q6; common consumer draws).

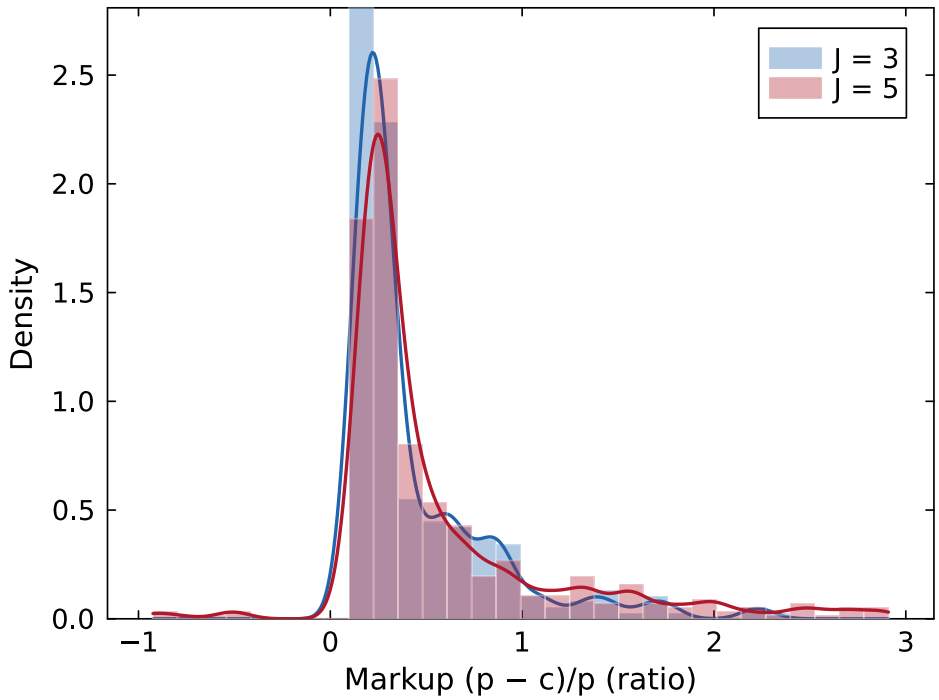


Figure 2: Distribution of markups, $J = 3$ versus $J = 5$ (markups below -1 excluded; axis truncated at the 95th percentile).

Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` (`pset_io`, Q6; common consumer draws).

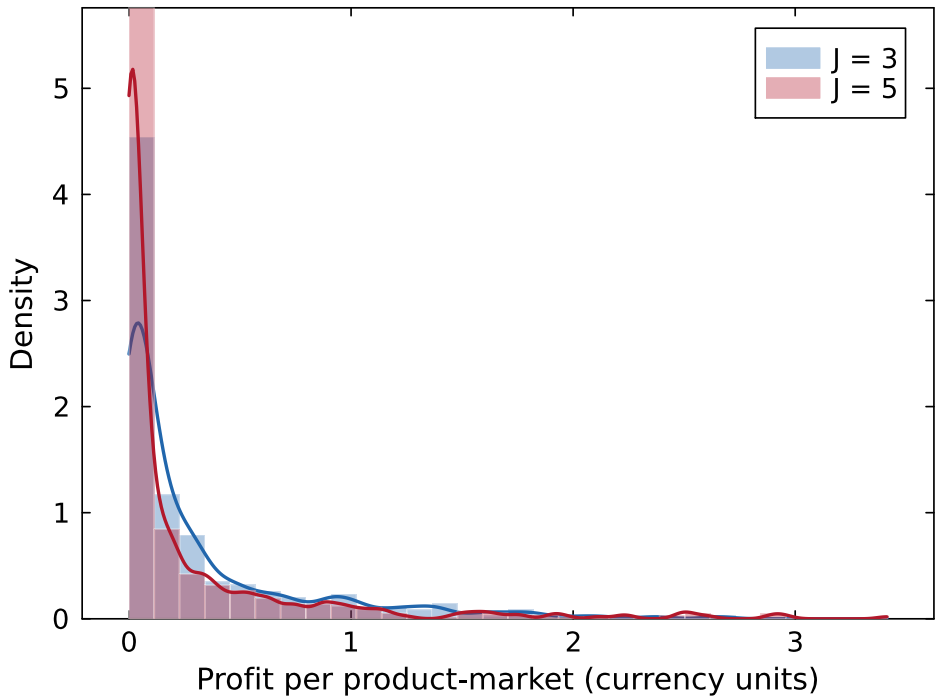


Figure 3: Distribution of profits, $J = 3$ versus $J = 5$.

Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` (`pset_io`, Q6; common consumer draws).

Table 1: Prices, markups and profits: $J = 3$ versus $J = 5$ (true parameters, product-market observations).

	$J = 3$					$J = 5$				
	Mean	SD	Median	Q25	Q75	Mean	SD	Median	Q25	Q75
Price	3.027	1.371	2.828	1.988	3.964	1.891	1.435	1.816	0.780	2.830
Markup	0.472	0.634	0.274	0.197	0.574	1.251	3.595	0.334	0.234	0.844
Profit	0.309	0.463	0.105	0.021	0.353	0.254	0.514	0.041	0.004	0.225

Pooled over products and markets (300 observations for $J = 3$, 500 for $J = 5$; 475 markups for $J = 5$). Markup is $(p - c)/p$. 25 $J = 5$ observations with markup < -1 are excluded from the markup row and from the markup figure. The markup figure also truncates the x-axis at the 95th percentile.

Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` (`pset_io`, Q6; common consumer draws).

6b

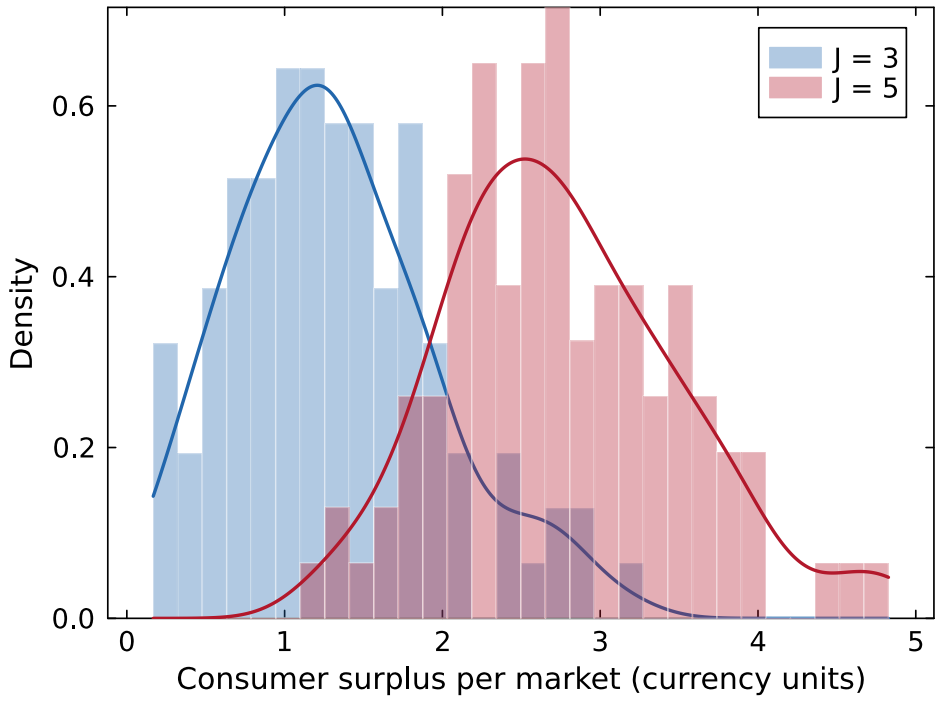


Figure 4: Distribution of consumer surplus, $J = 3$ versus $J = 5$.

Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` (`pset_io`, Q6; common consumer draws).

Table 2: Consumer surplus per market: $J = 3$ versus $J = 5$ (true parameters).

	$J = 3$					$J = 5$				
	Mean	SD	Median	Q25	Q75	Mean	SD	Median	Q25	Q75
Consumer surplus	1.314	0.642	1.251	0.851	1.729	2.765	0.747	2.697	2.258	3.228

100 markets per dataset; both datasets use the same $R = 500$ consumer draws.

Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` (`pset_io`, Q6; common consumer draws).

6c

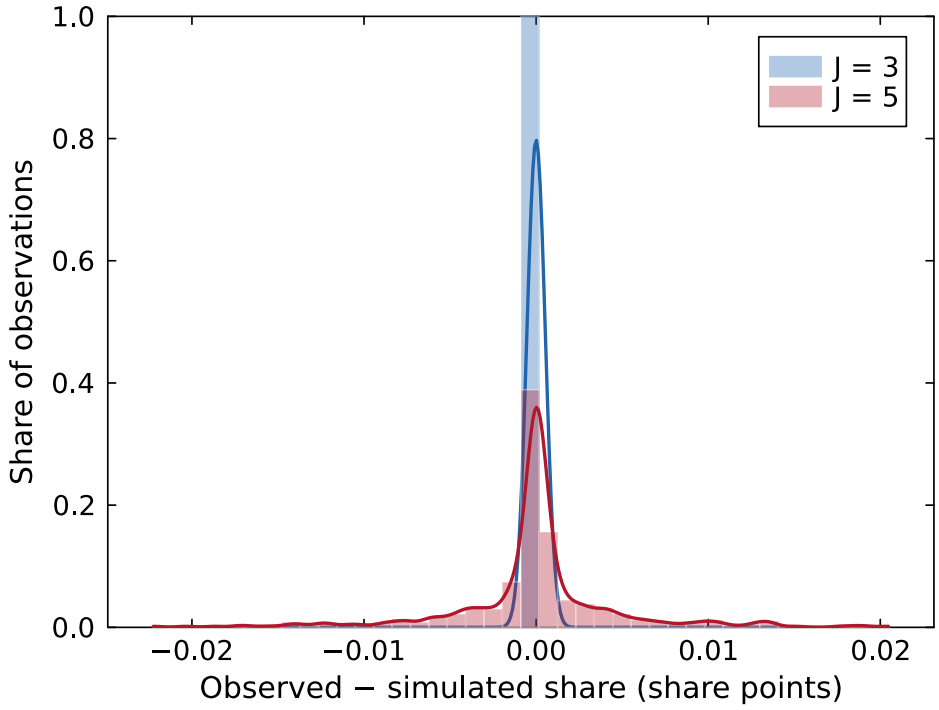


Figure 5: Observed minus simulated shares, $J = 3$ and $J = 5$.

Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` (`pset_io`, Q6; common consumer draws).

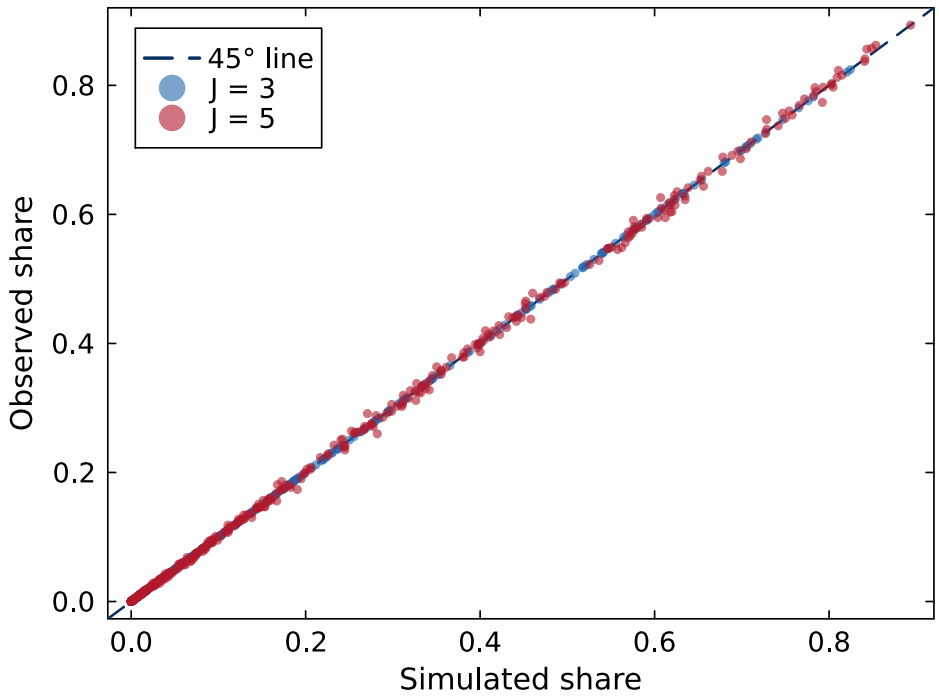


Figure 6: Observed versus simulated shares, $J = 3$ and $J = 5$.

Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` (`pset_io`, Q6; common consumer draws).

Table 3: Share discrepancies and adding-up checks (observed versus simulated at the true parameters).

	Mean	SD	Median	Q25	Q75	Max abs.
$J = 3$						
Inside: observed − simulated	-3.01×10^{-18}	7.24×10^{-17}	0	-1.39×10^{-17}	6.94×10^{-18}	3.89×10^{-16}
Outside: observed − simulated	2.71×10^{-17}	3.38×10^{-16}	0	-1.60×10^{-16}	2.22×10^{-16}	1.05×10^{-15}
$\sum_j s_{jt} + s_{0t} - 1$, observed	0	0	0	0	0	0
$\sum_j \hat{s}_{jt} + \hat{s}_{0t} - 1$, simulated	-2.11×10^{-17}	3.42×10^{-16}	0	-2.22×10^{-16}	2.22×10^{-16}	9.99×10^{-16}
$J = 5$						
Inside: observed − simulated	6.42×10^{-6}	4.81×10^{-3}	6.76×10^{-6}	-7.79×10^{-4}	8.21×10^{-4}	2.23×10^{-2}
Outside: observed − simulated	-3.21×10^{-5}	5.55×10^{-3}	-1.10×10^{-5}	-1.19×10^{-3}	7.30×10^{-4}	2.10×10^{-2}
$\sum_j s_{jt} + s_{0t} - 1$, observed	0	0	0	0	0	0
$\sum_j \hat{s}_{jt} + \hat{s}_{0t} - 1$, simulated	1.22×10^{-17}	4.91×10^{-16}	0	-3.33×10^{-16}	2.22×10^{-16}	1.33×10^{-15}

Inside rows: product-market observations; outside and adding-up rows: markets. Simulated shares average the logit choice probabilities over the $R = 500$ draws; the outside share is $1/(1 + \sum_j e^{u_{ijt}})$ averaged the same way. The observed outside share is $1 - \sum_j s_{jt}$ by construction, so its adding-up row is zero up to rounding.

Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` (`pset_io`, Q6; common consumer draws).

Question 7: demand moments and instruments

7a

Table 4: Sample moments $\mathrm{E}[\xi_{jt} z_{jt}]$ with characteristics, price and the Hausman price.

	Mean	SD	t-stat	p-value
$\mathrm{E}[\xi_{jt} \cdot 1]$	0.043	1.004	0.750	0.454
$\mathrm{E}[\xi_{jt} x_{2jt}]$	0.021	0.582	0.617	0.538
$\mathrm{E}[\xi_{jt} x_{3jt}]$	0.036	0.965	0.638	0.524
$\mathrm{E}[\xi_{jt} p_{jt}]$	0.295	3.670	1.392	0.165
$\mathrm{E}[\xi_{jt} \bar{p}_{j,-t}]$	0.144	3.169	0.785	0.433

Sample analogues over the $N = 300$ product-market observations of the $T = 100$, $J = 3$ dataset, using the true ξ_{jt}. SD is that of the observation-level product; $t = \bar{g}/(\mathrm{SD}/\sqrt{N})$ tests $H_0:$ mean $= 0$ (two-sided, t_{N-1}) treating observations as independent.

Source: $T = 100$, $J = 3$ simulation data with the true ξ_{jt} (`blp.compute_hausman`, `blp.compute_blp`).

7b

7c

Table 5: Sample moments $\mathrm{E}[\xi_{jt} z_{jt}]$ with BLP rival sums and the Hausman price.

	Mean	SD	t-stat	p-value
$\mathrm{E}[\xi_{jt} \sum_{k \neq j} x_{2kt}]$	0.046	1.154	0.686	0.493
$\mathrm{E}[\xi_{jt} \sum_{k \neq j} x_{3kt}]$	−0.009	1.230	−0.123	0.902
$\mathrm{E}[\xi_{jt} \bar{p}_{j,-t}]$	0.144	3.169	0.785	0.433

Sample analogues over the $N = 300$ product-market observations of the $T = 100$, $J = 3$ dataset, using the true ξ_{jt}. SD is that of the observation-level product; $t = \bar{g}/(\mathrm{SD}/\sqrt{N})$ tests $H_0:$ mean $= 0$ (two-sided, t_{N-1}) treating observations as independent. Own-firm sums equal x_{jt} itself (single-product firms), so they replicate the $\mathrm{E}[\xi_{jt} x_{jt}]$ rows of the previous table.

Source: $T = 100$, $J = 3$ simulation data with the true ξ_{jt} (`blp.compute_hausman`, `blp.compute_blp`).

Question 8: mixed logit by nested fixed point

8a

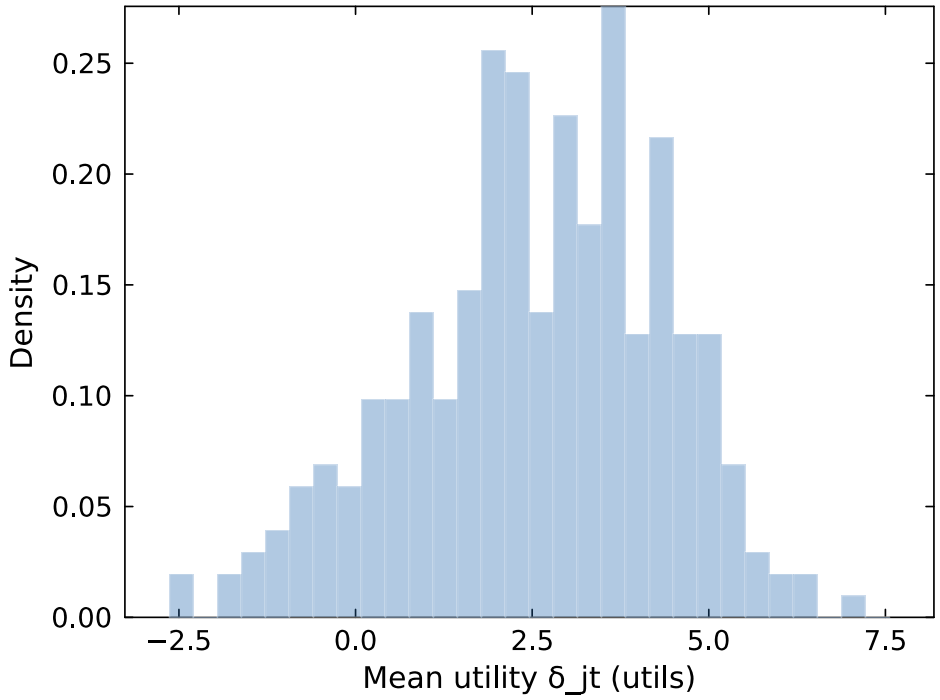


Figure 7: Distribution of mean utility δ_{jt} ($J = 3$).

Source: true-parameter simulation of `100markets3products.mat` ($J = 3$).

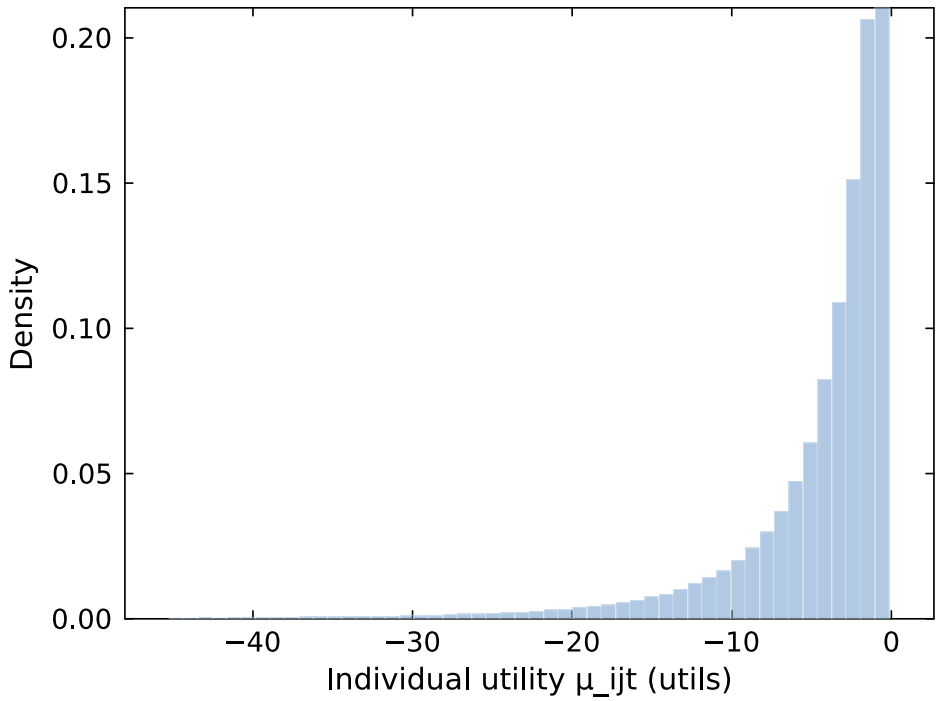


Figure 8: Distribution of individual utility deviations μ_{ijt} ($J = 3$; displayed between the 0.5th and 99.5th percentiles).

Source: true-parameter simulation of `100markets3products.mat` ($J = 3$).

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Table 6: Mixed-logit estimates by starting value ($T = 100$, $J = 3$, BLP instruments, no cost shifter).

	Truth	Start 1	Start 2†	Start 3	Bias
Intercept	5.000	4.981	4.981	4.981	−0.019
		(0.404)	(0.404)	(0.404)	
x_2	1.000	0.966	0.966	0.966	−0.034
		(0.225)	(0.225)	(0.225)	
x_3	1.000	1.036	1.036	1.036	0.036
		(0.076)	(0.076)	(0.076)	
Price coefficient $\bar{\alpha}$	1.000	0.971	0.971	0.971	−0.029
		(0.116)	(0.116)	(0.116)	
Price heterogeneity σ_α	1.000	1.006	1.006	1.006	0.006
		(0.080)	(0.080)	(0.080)	
Starting value σ_α^0		0.922	1.659	0.449	
GMM criterion		4.60×10^{-3}	4.60×10^{-3}	4.60×10^{-3}	
Converged (Optim)		yes	yes	yes	

Standard errors in parentheses (GMM sandwich, weighting $W = (Z'Z)^{-1}$). † marks the selected estimate, the start with the lowest GMM criterion; Bias = selected estimate minus truth. The price coefficient is reported as $\bar{\alpha}$ (utility $-\alpha_i p$), truth $\bar{\alpha} = 1$.

Source: estimated model `models/model.jdl2` ($T = 100$, $J = 3$, BLP instruments, three starting values).

Table 7: Convergence and contraction-mapping diagnostics by starting value ($T = 100$, $J = 3$).

Start	σ_α^0	$\hat{\sigma}_\alpha$	$\lvert\hat{\sigma}_\alpha - \hat{\sigma}_\alpha^*\rvert$	Criterion	Converged	Outer iter.	$\lVert\nabla\rVert_\infty$	Contr. iter. (10^{-3})	Contr. iter. (10^{-10})	Sup. error at $\hat{\delta}$
1	0.922	1.006	3.42×10^{-12}	4.60×10^{-3}	yes	4	1.37×10^{-11}	315	3814	6.28×10^{-11}
2†	1.659	1.006	0	4.60×10^{-3}	yes	4	9.33×10^{-10}	315	3814	5.73×10^{-11}
3	0.449	1.006	3.43×10^{-12}	4.60×10^{-3}	yes	4	1.27×10^{-11}	315	3814	6.28×10^{-11}

Optimiser: L-BFGS in a box $[0.1, 10]$ on σ_α (Optim, Fminbox). Contraction iterations count steps of $\delta \leftarrow \delta + \log s^{\text{obs}} - \log s(\delta; \sigma)$ from the logit starting value until the L2 norm of the update falls below the tolerance. Sup. error is $\max_{jt} \lvert \log s_{jt}^{\text{obs}} - \log s_{jt}(\hat{\delta}; \hat{\sigma}) \rvert$ at the saved $\hat{\delta}$. † marks the selected start.

Source: estimated model `models/model.jdl2` ($T = 100$, $J = 3$, BLP instruments, three starting values).

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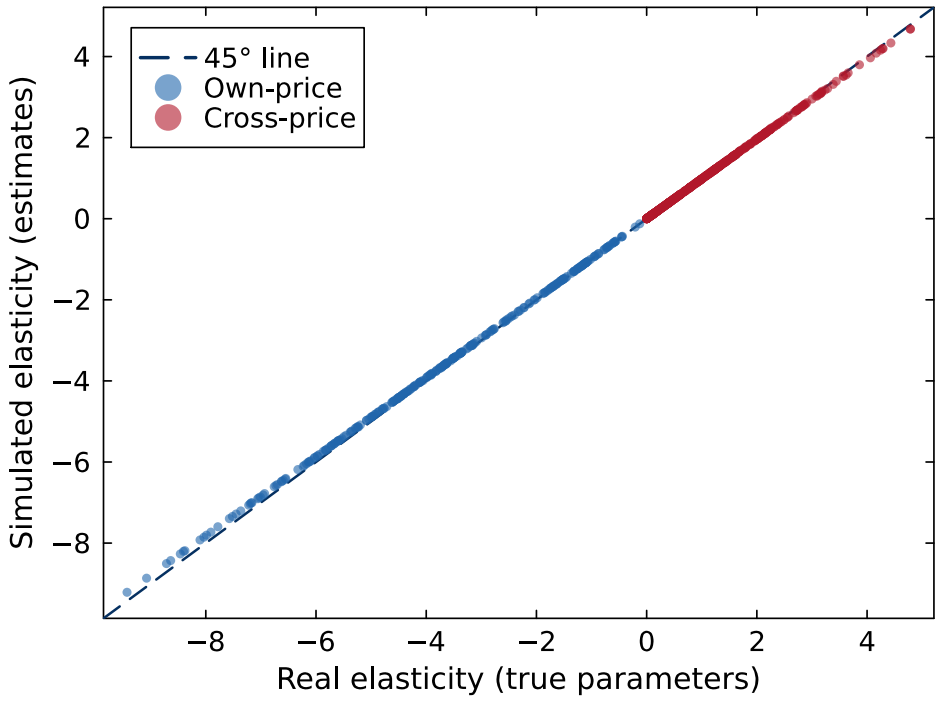


Figure 9: Simulated (at estimates) versus real elasticities: own- and cross-price.
Source: estimated model `models/model.jdl2` ($T = 100$, $J = 3$, BLP instruments, three starting values).

Table 8: Elasticity of product 1's share with respect to the price of product 1: real versus simulated ($T = 100$ markets).

	Mean	SD	Median	Q25	Q75
Real	−5.097	1.938	−5.008	−6.384	−3.850
Simulated	−4.991	1.889	−4.908	−6.241	−3.770

Element $(1, 1)$ of the elasticity matrix $\partial s_j / \partial p_k \cdot p_k / s_j$ across the 100 markets; simulated uses the Q8c estimates, real uses the true parameters.

Source: estimated model `models/model.jdl2` ($T = 100$, $J = 3$, BLP instruments, three starting values).

Table 9: Elasticity of product 1's share with respect to the price of product 2: real versus simulated ($T = 100$ markets).

	Mean	SD	Median	Q25	Q75
Real	1.395	1.013	1.237	0.546	2.139
Simulated	1.367	0.992	1.215	0.537	2.096

Element $(1, 2)$ of the elasticity matrix $\partial s_j / \partial p_k \cdot p_k / s_j$ across the 100 markets; simulated uses the Q8c estimates, real uses the true parameters.

Source: estimated model `models/model.jdl2` ($T = 100$, $J = 3$, BLP instruments, three starting values).

Table 10: Elasticity of product 1's share with respect to the price of product 3: real versus simulated ($T = 100$ markets).

	Mean	SD	Median	Q25	Q75
Real	1.000	0.884	0.896	0.310	1.445
Simulated	0.980	0.867	0.879	0.304	1.418

Element $(1, 3)$ of the elasticity matrix $\partial s_j / \partial p_k \cdot p_k / s_j$ across the 100 markets; simulated uses the Q8c estimates, real uses the true parameters.

Source: estimated model `models/model.jdl2` ($T = 100$, $J = 3$, BLP instruments, three starting values).

Table 11: Elasticity of product 2's share with respect to the price of product 1: real versus simulated ($T = 100$ markets).

	Mean	SD	Median	Q25	Q75
Real	0.593	0.696	0.327	0.091	0.925
Simulated	0.582	0.683	0.319	0.089	0.907

Element $(2, 1)$ of the elasticity matrix $\partial s_j / \partial p_k \cdot p_k / s_j$ across the 100 markets; simulated uses the Q8c estimates, real uses the true parameters.

Source: estimated model `models/model.jdl2` ($T = 100$, $J = 3$, BLP instruments, three starting values).

Table 12: Elasticity of product 2's share with respect to the price of product 2: real versus simulated ($T = 100$ markets).

	Mean	SD	Median	Q25	Q75
Real	−2.622	1.666	−2.280	−3.693	−1.207
Simulated	−2.573	1.630	−2.238	−3.623	−1.186

Element $(2, 2)$ of the elasticity matrix $\partial s_j / \partial p_k \cdot p_k / s_j$ across the 100 markets; simulated uses the Q8c estimates, real uses the true parameters.

Source: estimated model `models/model.jdl2` ($T = 100$, $J = 3$, BLP instruments, three starting values).

Table 13: Elasticity of product 2's share with respect to the price of product 3: real versus simulated ($T = 100$ markets).

	Mean	SD	Median	Q25	Q75
Real	0.985	0.917	0.768	0.282	1.330
Simulated	0.968	0.900	0.752	0.277	1.305

Element $(2, 3)$ of the elasticity matrix $\partial s_j / \partial p_k \cdot p_k / s_j$ across the 100 markets; simulated uses the Q8c estimates, real uses the true parameters.

Source: estimated model `models/model.jdl2` ($T = 100$, $J = 3$, BLP instruments, three starting values).

Table 14: Elasticity of product 3's share with respect to the price of product 1: real versus simulated ($T = 100$ markets).

	Mean	SD	Median	Q25	Q75
Real	0.602	0.700	0.324	0.098	0.937
Simulated	0.591	0.687	0.318	0.096	0.920

Element $(3, 1)$ of the elasticity matrix $\partial s_j / \partial p_k \cdot p_k / s_j$ across the 100 markets; simulated uses the Q8c estimates, real uses the true parameters.

Source: estimated model `models/model.jdl2` ($T = 100$, $J = 3$, BLP instruments, three starting values).

Table 15: Elasticity of product 3's share with respect to the price of product 2: real versus simulated ($T = 100$ markets).

	Mean	SD	Median	Q25	Q75
Real	1.441	1.061	1.253	0.514	2.136
Simulated	1.415	1.041	1.231	0.504	2.091

Element $(3, 2)$ of the elasticity matrix $\partial s_j / \partial p_k \cdot p_k / s_j$ across the 100 markets; simulated uses the Q8c estimates, real uses the true parameters.

Source: estimated model `models/model.jdl2` ($T = 100$, $J = 3$, BLP instruments, three starting values).

Table 16: Elasticity of product 3's share with respect to the price of product 3: real versus simulated ($T = 100$ markets).

	Mean	SD	Median	Q25	Q75
Real	−3.407	1.819	−3.488	−4.497	−1.771
Simulated	−3.343	1.778	−3.422	−4.411	−1.742

Element $(3, 3)$ of the elasticity matrix $\partial s_j / \partial p_k \cdot p_k / s_j$ across the 100 markets; simulated uses the Q8c estimates, real uses the true parameters.

Source: estimated model `models/model.jdl2` ($T = 100$, $J = 3$, BLP instruments, three starting values).

8e

Table 17: Mixed-logit estimates by starting value ($T = 10$, $J = 3$, BLP instruments, no cost shifter).

	Truth	Start 1†	Bias
Intercept	5.000	4.101	−0.899
		(1.235)	
x_2	1.000	0.495	−0.505
		(0.903)	
x_3	1.000	1.012	0.012
		(0.331)	
Price coefficient $\bar{\alpha}$	1.000	0.993	−0.007
		(0.330)	
Price heterogeneity σ_α	1.000	0.100	−0.900
		(1.945)	
Starting value σ_α^0		0.922	
GMM criterion		2.89×10^{-1}	
Converged (Optim)		yes	

Standard errors in parentheses (GMM sandwich, weighting $W = (Z'Z)^{-1}$). † marks the selected estimate, the start with the lowest GMM criterion; Bias = selected estimate minus truth. The price coefficient is reported as $\bar{\alpha}$ (utility $-\alpha_i p$), truth $\bar{\alpha} = 1$. Only one starting value was run for this model.

Source: estimated model `models/model_ls.jdl2` ($T = 10$, $J = 3$).

Table 18: Convergence and contraction-mapping diagnostics by starting value ($T = 10$, $J = 3$).

Start	σ_α^0	$\hat{\sigma}_\alpha$	$\lvert\hat{\sigma}_\alpha - \hat{\sigma}_\alpha^*\rvert$	Criterion	Converged	Outer iter.	$\lVert\nabla\rVert_\infty$	Contr. iter. (10^{-3})	Contr. iter. (10^{-10})	Sup. error at $\hat{\delta}$
1†	0.922	0.100	0	2.89×10^{-1}	yes	3	3.85×10^{-9}	49	10000	1.71×10^{-6}

Optimiser: L-BFGS in a box $[0.1, 10]$ on σ_α (Optim, Fminbox). Contraction iterations count steps of $\delta \leftarrow \delta + \log s^{\text{obs}} - \log s(\delta; \sigma)$ from the logit starting value until the L2 norm of the update falls below the tolerance. Sup. error is $\max_{jt} \lvert \log s_{jt}^{\text{obs}} - \log s_{jt}(\hat{\delta}; \hat{\sigma})\rvert$ at the saved $\hat{\delta}$. † marks the selected start. Only one starting value was run for this model.

Source: estimated model `models/model_ls.jdl2` ($T = 10$, $J = 3$).

8f

Table 19: Price as its own instrument versus IV estimates.

	Truth	IV (Q8c)	Price as instrument
Intercept	5.000	4.981	4.865
		(0.404)	(0.341)
x_2	1.000	0.966	0.944
		(0.225)	(0.220)
x_3	1.000	1.036	1.036
		(0.076)	(0.076)
Price coefficient $\bar{\alpha}$	1.000	0.971	0.924
		(0.116)	(0.082)
Price heterogeneity σ_α	1.000	1.006	1.014
		(0.080)	(0.080)
Implied objects (mean over product-markets)			
Own-price elasticity	−3.709	−3.636	−3.518
Cross-price elasticity	1.003	0.984	0.953
Markup $(p-c)/p$	0.472	0.479	0.492

$T = 100$, $J = 3$. Standard errors in parentheses. Truth column for implied objects uses the true parameters; markups are model-implied under Bertrand pricing.

Source: estimated models `models/model_inc.jdl2` (price as instrument) and `models/model.jdl2`.

Question 9: cost shifters and the supply side

9a

Table 20: Demand estimates without and with the cost-shifter instrument w_j ($T = 100$, $J = 3$).

	Truth	BLP instr. (Q8c)	+ cost shifter w_j
Intercept	5.000	4.981	5.003
		(0.404)	(0.397)
x_2	1.000	0.966	0.971
		(0.225)	(0.226)
x_3	1.000	1.036	1.037
		(0.076)	(0.076)
Price coefficient $\bar{\alpha}$	1.000	0.971	0.980
		(0.116)	(0.113)
Price heterogeneity σ_α	1.000	1.006	1.004
		(0.080)	(0.080)
Implied objects (mean over product-markets)			
Own-price elasticity	−3.709	−3.636	−3.658
Profit per product-market	0.309	0.314	0.312
Consumer surplus per market	1.314	1.346	1.336

Standard errors in parentheses. Truth column for implied objects: true parameters (profit uses the true marginal cost). Model-implied profits use $(p - \hat{c})s$ with $\hat{c}$ recovered under Bertrand pricing.

Source: estimated models `models/model_ss.jdl2` (+ cost shifter) and `models/model.jdl2`.

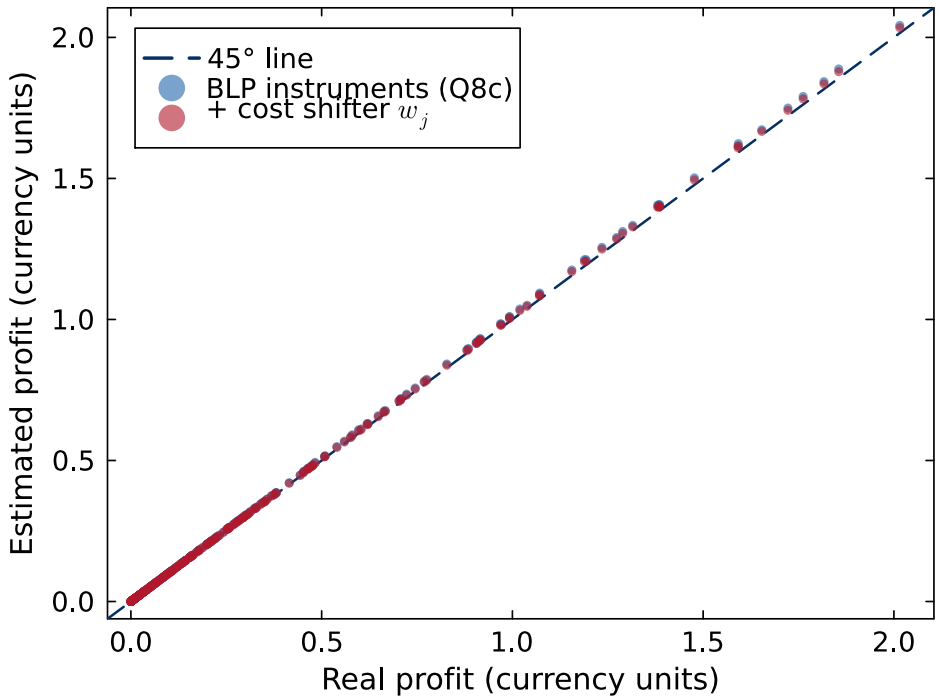


Figure 10: Profits at the estimates versus real profits, by specification.

Source: estimated models `models/model_ss.jdl2` (+ cost shifter) and `models/model.jdl2`.

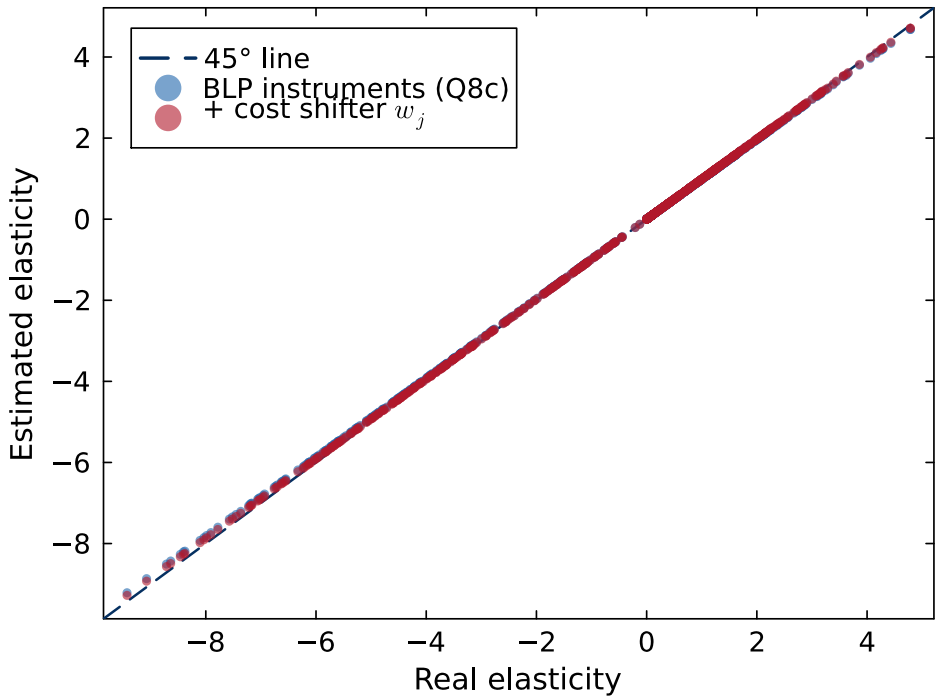


Figure 11: Elasticities at the estimates versus real elasticities, by specification.

Source: estimated models `models/model_ss.jdl2` (+ cost shifter) and `models/model.jdl2`.

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9d

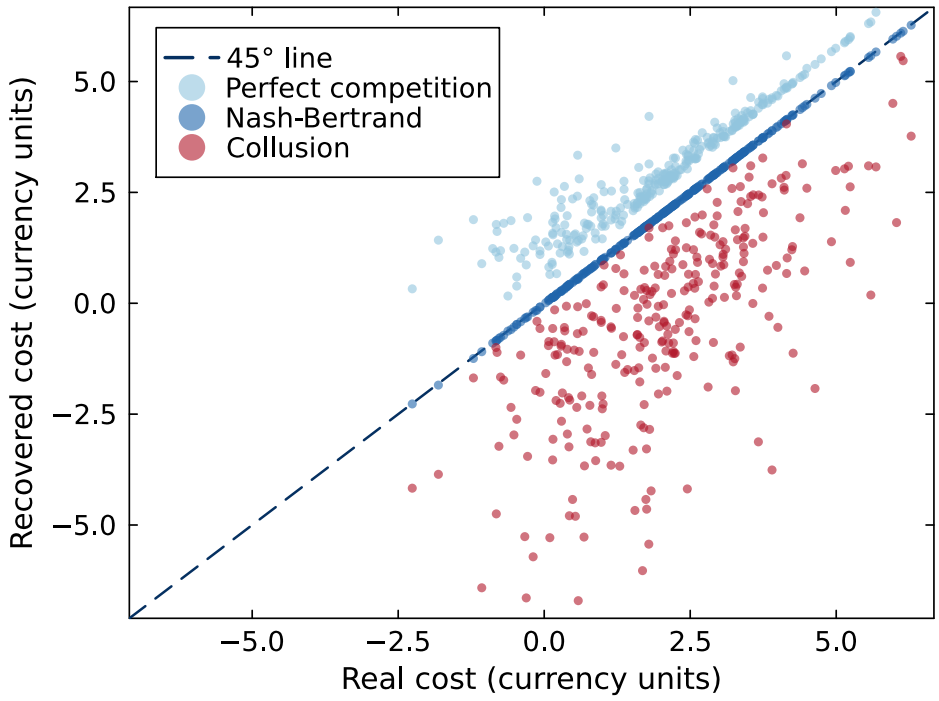


Figure 12: Recovered versus real marginal cost under perfect competition, Nash-Bertrand and collusion (axes truncated at the 0.5th and 99.5th percentiles).

Source: demand estimates `models/model_ss.jdl2`; marginal cost recovered under three ownership matrices.

9e

Table 21: Joint demand and supply estimates under three conduct assumptions ($T = 100$, $J = 3$).

	Truth	Nash-Bertrand	Perfect competition	Collusion
Intercept	5.000	5.017	5.003	3.995
		(0.397)	(0.397)	(1.066)
x_2	1.000	0.974	0.971	0.759
		(0.226)	(0.226)	(0.266)
x_3	1.000	1.039	1.037	0.880
		(0.076)	(0.076)	(0.146)
Price coefficient $\bar{\alpha}$	1.000	0.983	0.980	0.754
		(0.113)	(0.113)	(0.278)
Cost intercept γ_0	2.000	1.940	2.951	−1.174
		(0.074)	(0.055)	(0.417)
Cost on z_{jt} (γ_2)	1.000	0.916	0.790	0.581
		(0.056)	(0.051)	(0.585)
Cost on w_j (γ_1)	1.000	0.979	0.776	0.627
		(0.075)	(0.069)	(0.715)
Price heterogeneity σ_α	1.000	1.009	1.004	0.659
		(0.080)	(0.080)	(0.355)

Standard errors in parentheses (joint GMM sandwich with demand moments for ξ_{jt} and supply moments for ω_{jt}). Estimates are re-evaluated at the saved $\hat{\sigma}_\alpha$ (see the note in the notebook). Under perfect competition marginal cost equals price, so supply moments do not depend on σ_α.

Source: joint estimates `models/model_system_{bert,comp,coll}.jdl2`, re-evaluated at the saved $\hat{\sigma}_\alpha$.

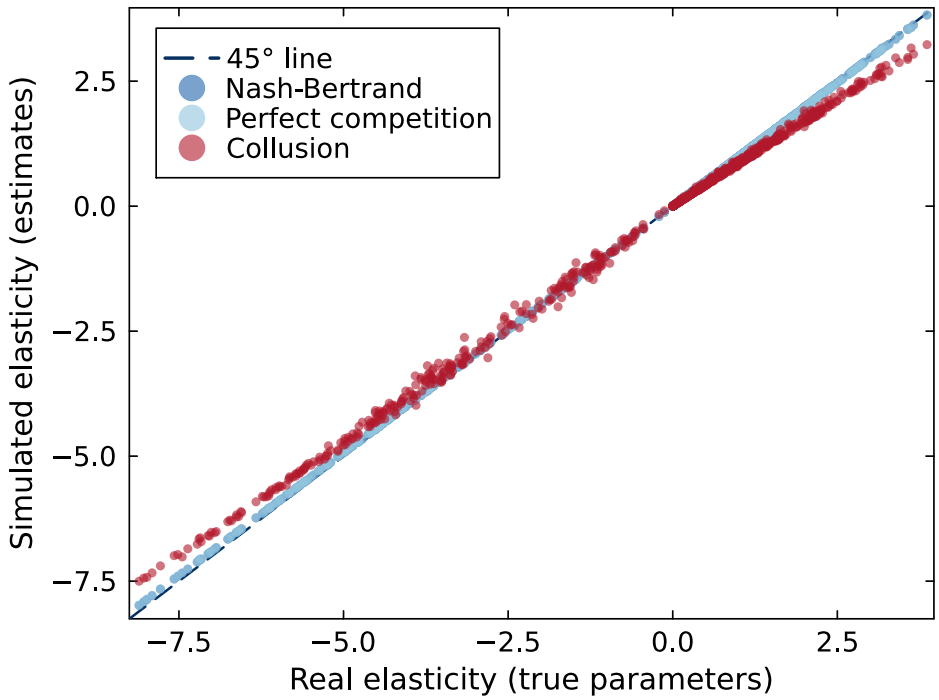


Figure 13: Simulated versus real elasticities by conduct assumption (axes truncated at the 1st and 99th percentiles).

Source: joint estimates `models/model_system_{bert,comp,coll}.jdl2`, re-evaluated at the saved $\hat{\sigma}_\alpha$.

Table 22: Elasticities under each conduct assumption versus the true elasticities.

	Mean	SD	Median	Q25	Q75
Real: own-price	−3.709	2.080	−3.650	−5.084	−1.744
Real: cross-price	1.003	0.947	0.771	0.210	1.475
Nash-Bertrand: own-price	−3.661	2.049	−3.613	−5.011	−1.722
Nash-Bertrand: cross-price	0.991	0.936	0.761	0.207	1.459
Perfect competition: own-price	−3.658	2.046	−3.609	−5.007	−1.722
Perfect competition: cross-price	0.989	0.935	0.760	0.207	1.456
Collusion: own-price	−3.505	1.924	−3.430	−4.814	−1.713
Collusion: cross-price	0.860	0.807	0.668	0.181	1.283

Own- and cross-price elasticities pooled over the 100 markets and 3 products (300 own, 600 cross).

Source: joint estimates `models/model_system_{bert,comp,coll}.jdl2`, re-evaluated at the saved $\hat{\sigma}_\alpha$.

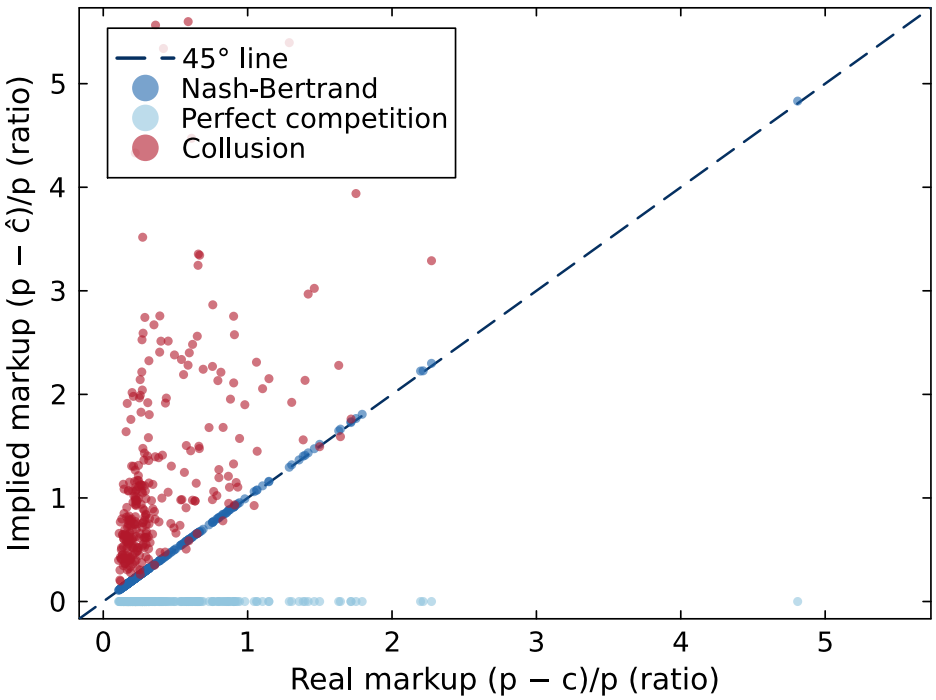


Figure 14: Implied versus real markups by conduct assumption (axes truncated at the 1st and 99th percentiles).

Source: joint estimates `models/model_system_{bert,comp,coll}.jdl2`, re-evaluated at the saved $\hat{\sigma}_\alpha$.

Table 23: Markups under each conduct assumption versus the true markups.

	Mean	SD	Median	Q25	Q75
Real	0.472	0.634	0.274	0.197	0.574
Nash-Bertrand	0.477	0.636	0.277	0.200	0.581
Perfect competition	0.000	0.000	0.000	0.000	0.000
Collusion	4.425	37.392	0.971	0.616	1.680

Markup $(p - c)/p$ over the 300 product-market observations; implied markups use the cost recovered from the first-order condition under each conduct assumption.

Source: joint estimates `models/model_system_{bert,comp,coll}.jdl2`, re-evaluated at the saved $\hat{\sigma}_\alpha$.

Question 10: merger simulation

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10b

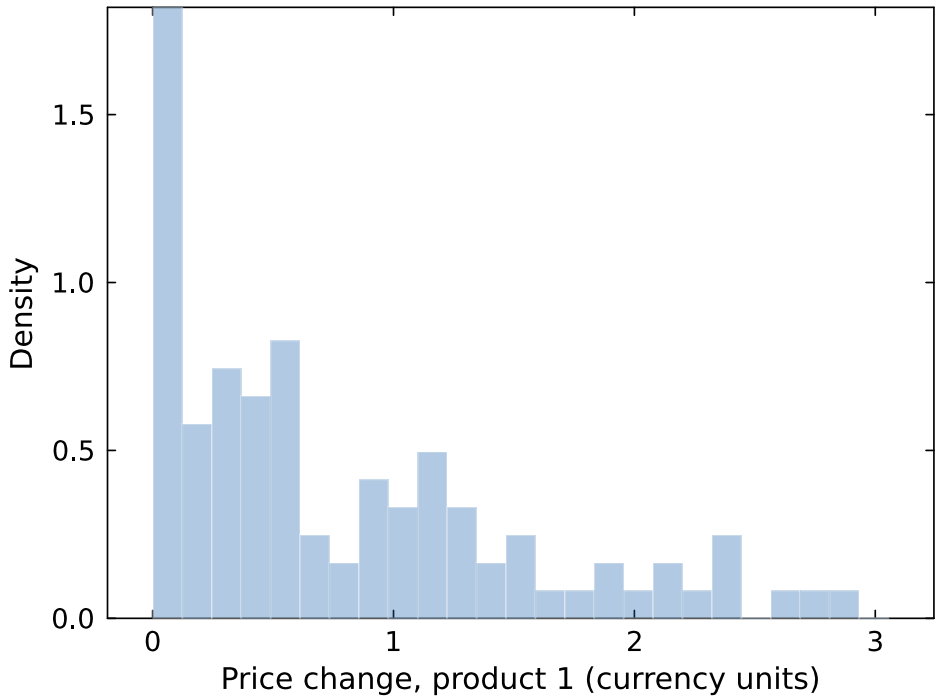


Figure 15: Post-merger price change, product 1 (merging).

Source: merger simulation of firms 1 and 2 from `models/model_system_bert.jdl2`.

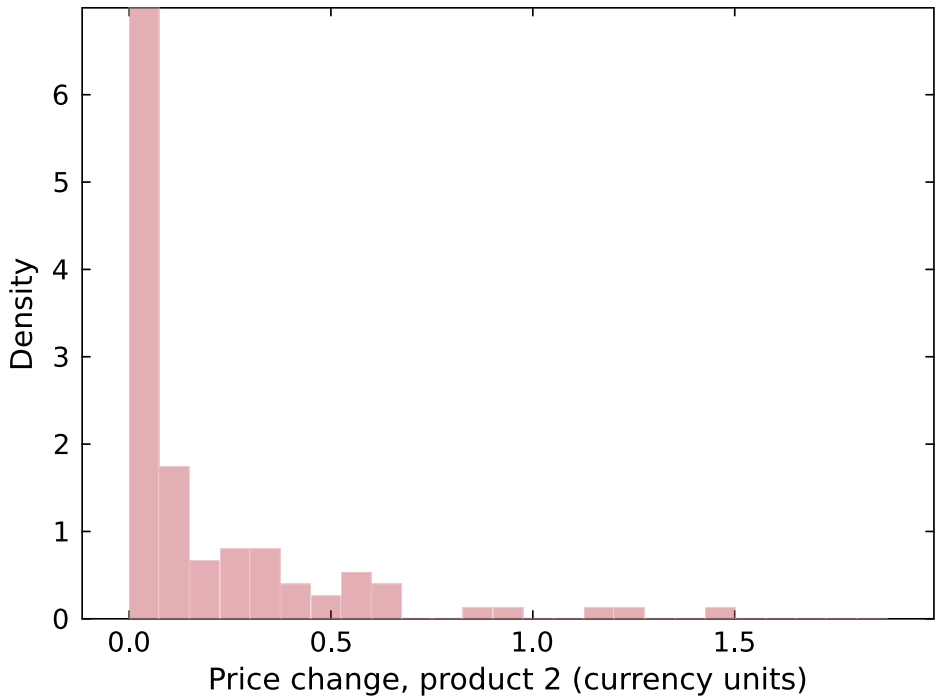


Figure 16: Post-merger price change, product 2 (merging).

Source: merger simulation of firms 1 and 2 from `models/model_system_bert.jdl2`.

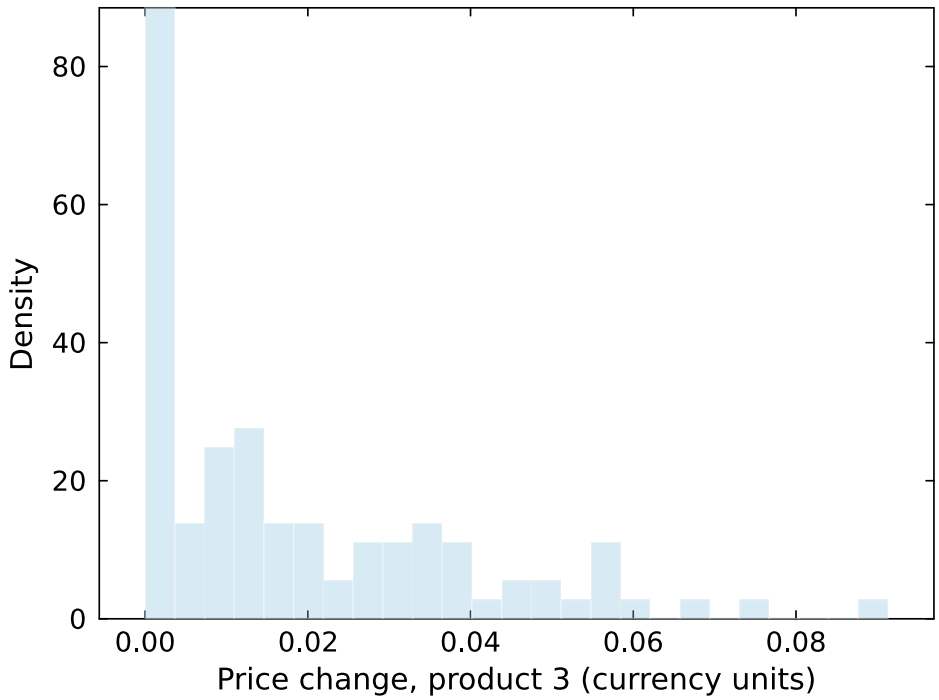


Figure 17: Post-merger price change, product 3 (non-merging).

Source: merger simulation of firms 1 and 2 from `models/model_system_bert.jdl2`.

Table 24: Post-merger price changes by product (firms 1 and 2 merge).

	Mean	SD	Median	Q25	Q75
Product 1 (merging)	0.800	0.771	0.539	0.170	1.204
Product 2 (merging)	0.213	0.329	0.075	0.017	0.270
Product 3 (non-merging)	0.019	0.021	0.012	0.002	0.032

Post-merger minus pre-merger equilibrium price, over the 100 markets (currency units).

Source: merger simulation of firms 1 and 2 from `models/model_system_bert.jdl2`.

Table 25: Change in consumer, producer and total surplus after the merger.

	Mean	SD	Median	Q25	Q75
Consumer surplus	−0.084	0.147	−0.029	−0.092	−0.008
Producer surplus	0.045	0.078	0.017	0.004	0.048
Total surplus	−0.039	0.074	−0.009	−0.043	−0.002

Per market (100 markets, currency units). Producer surplus is the sum of product profits $(p - \hat{c})s$; total surplus is the sum of consumer and producer surplus.

Source: merger simulation of firms 1 and 2 from `models/model_system_bert.jdl2`.

10c

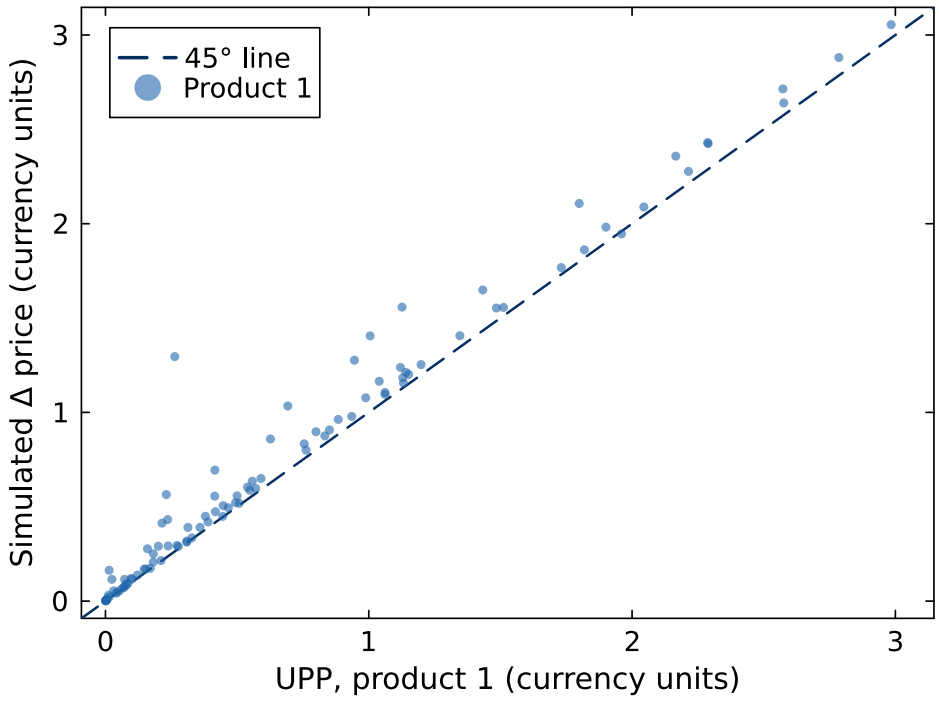


Figure 18: Upward pricing pressure versus simulated price change, product 1.
Source: merger simulation of firms 1 and 2 from `models/model_system_bert.jdl2`.

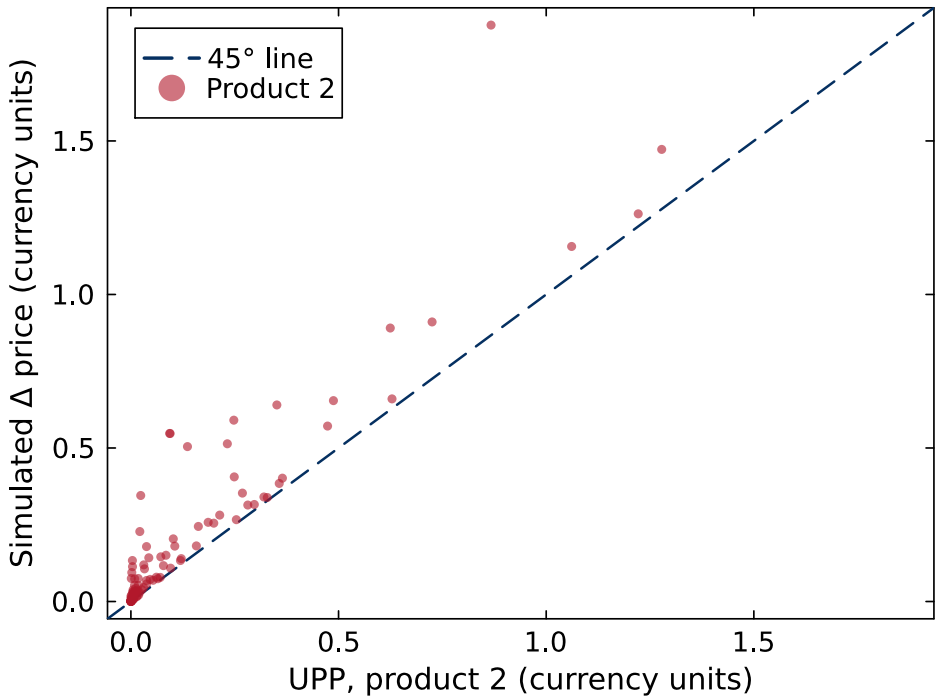


Figure 19: Upward pricing pressure versus simulated price change, product 2.
Source: merger simulation of firms 1 and 2 from `models/model_system_bert.jdl2`.

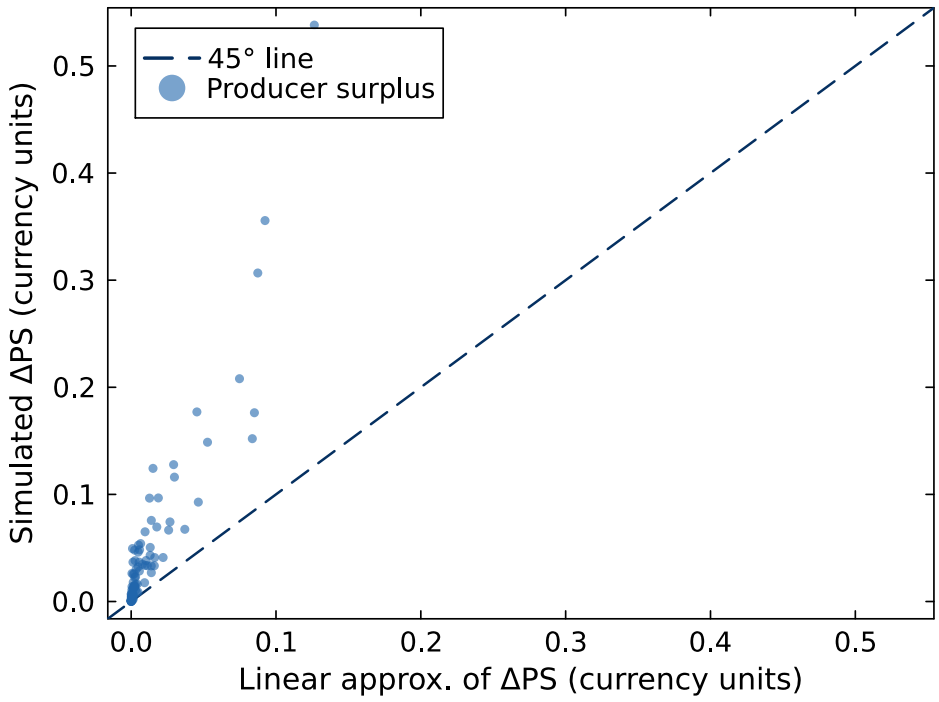


Figure 20: Linear approximation versus simulated change in producer surplus.
Source: merger simulation of firms 1 and 2 from `models/model_system_bert.jdl2`.

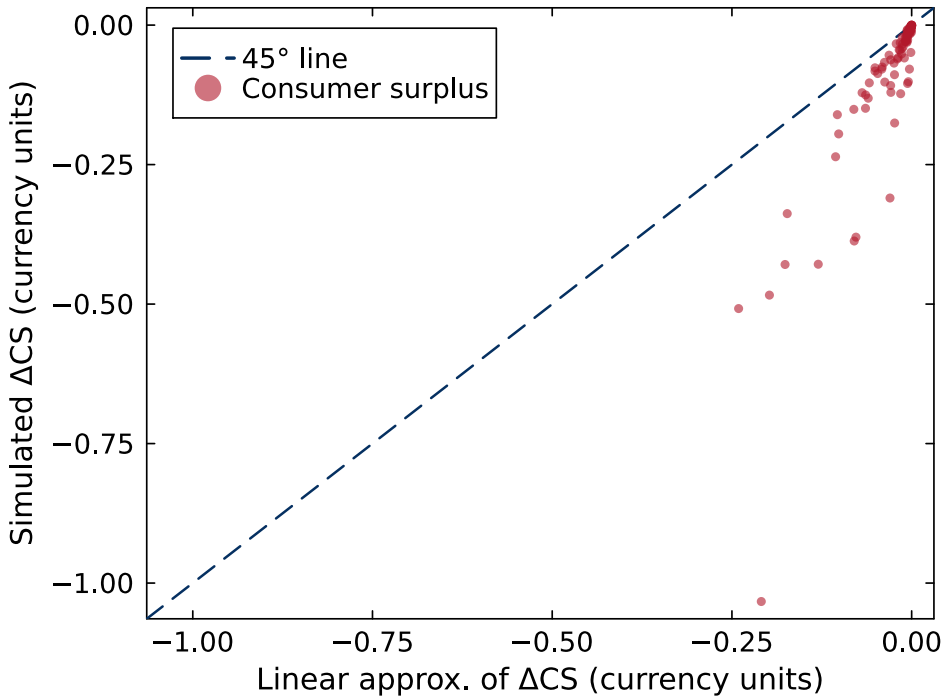


Figure 21: Linear approximation versus simulated change in consumer surplus.
Source: merger simulation of firms 1 and 2 from `models/model_system_bert.jdl2`.

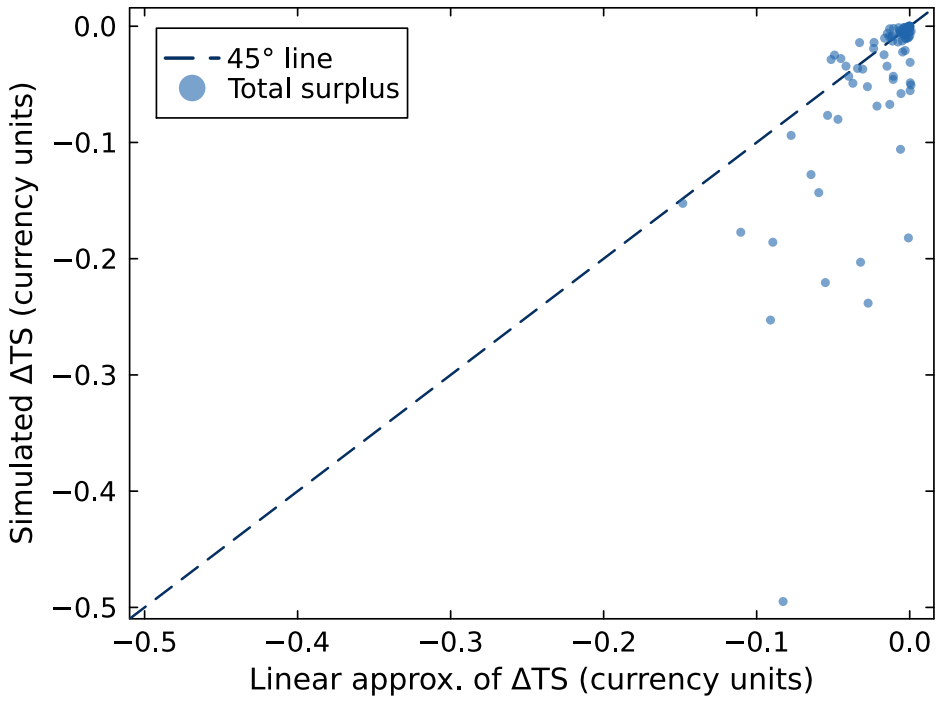


Figure 22: Linear approximation versus simulated change in total surplus.
Source: merger simulation of firms 1 and 2 from `models/model_system_bert.jdl2`.